

travel booking

Software Requirements Specification

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The system will run on a server equipped with at least 16GB RAM and a dual-core processor running at 2GHz. It requires a minimum storage capacity of 500GB.

The application needs to support Windows, Linux, and macOS operating systems. It shall have a responsive web interface accessible via HTTP/HTTPS protocols. The system must integrate with third-party APIs for flight and hotel information retrieval.

The system communicates over HTTPS using RESTful API endpoints. It supports JSON format for data exchange between client and server.

Usability: The system shall provide an intuitive user interface that is easy to navigate and understand.

Reliability: The system shall ensure data integrity through validation checks and error handling mechanisms.

Performance: The system shall process requests within 2 seconds under normal load conditions.

Portability: The system shall be deployable across different platforms without significant modification.

INTRODUCTION

Purpose: To define the system requirements for a travel booking application.

Scope: This document covers the requirements for a travel booking system including search, reservation, payment, and customer service functionalities.

Definitions/Acronyms: UI

User Interface; APIApplication Programming InterfaceReferences: IEEE Std 830-1998

Overview: This document outlines the necessary requirements for developing a comprehensive travel booking solution.

OVERALL DESCRIPTION

Product Perspective: The product aims to facilitate seamless travel planning and booking experiences for users worldwide.

Product Functions: Users can search for flights and hotels, compare prices, make reservations, pay online, manage bookings, and access customer support services.

User Characteristics: Target users include frequent travelers, business professionals, and leisure tourists who require efficient and reliable booking processes.

Constraints: The system must operate efficiently on standard hardware configurations and comply with industry standards for data privacy and security.

Assumptions/Dependencies: The system relies on third-party providers for real-time flight and hotel information updates.

SPECIFIC REQUIREMENTS

External Interface Requirements

User Interface: The system provides a graphical user interface allowing users to interact with various functions such as searching, comparing, reserving, and managing their bookings.

Hardware Interface: The system runs on servers meeting specified hardware requirements.

Software Interface: The application supports Windows, Linux, and macOS operating systems.

Communication Interface: Data exchanges occur over secure HTTPS connections using RESTful API endpoints supporting JSON format.

Functional Requirements

FR-01 Search Multi-City Itinerary

Input: User enters origin, destination cities, dates, number of passengers, preferred airlines/hotels.

Processing: System retrieves relevant flight and hotel details from third-party sources.

Output: Displayed results include available itineraries with corresponding costs, baggage allowances, and refund policies.

Priority: High

FR-02 Compare Fares and Policies

Input: Selected itineraries from previous step.

Processing: System compares pricing and policy differences among chosen options.

Output: Results show detailed comparisons highlighting cost variations and specific terms.

Priority: Medium

FR-03 Reserve Seats and Add Insurance

Input: Chosen itinerary details.

Processing: System reserves selected seats and adds optional insurance coverage if requested.

Output: Confirmation message includes seat numbers, reservation ID, and insurance details.

Priority: High

FR-04 Receive E-Ticket and Store Passenger Profile

Input: Reservation confirmation.

Processing: System generates electronic tickets and stores passenger profile data securely.

Output: User receives email containing digital ticket and access to stored profile for future reference.

Priority: Low

FR-05 Handle Voucher Redemption

Input: Valid voucher code provided by user.

Processing: System verifies voucher validity and applies discount to current booking.

Output: Updated total amount displayed along with applicable discounts.

Priority: Medium

FR-06 Manage Fare Changes and Alternative Options

Input: Current booking status and updated preferences.

Processing: System evaluates new options based on changed criteria and presents alternatives.

Output: New set of possible itineraries and associated costs presented to user.

Priority: High

Non-functional Requirements

Usability: The system ensures ease of navigation and understanding throughout all interfaces.

Reliability: Data integrity maintained through robust validation and error handling procedures.

Performance: Requests processed within 2 seconds under typical usage scenarios.

Portability: Deployment feasible across diverse environments requiring minimal adjustments.

SYSTEM FEATURES

Feature 1: Booking Engine

Enables users to find and book flights and accommodations simultaneously. Supports dynamic pricing and real-time availability checking. Facilitates comparison shopping by presenting side-by-side price and feature contrasts.

Feature 2: Payment Gateway Integration
Processes transactions securely using encrypted channels. Accepts multiple payment methods including credit cards, debit cards, and digital wallets. Provides receipt generation upon successful transaction completion.

Feature 3: Customer Support Portal

Offers live chat functionality for immediate assistance. Includes FAQ section addressing common queries about bookings and cancellations. Maintains historical records of interactions for reference purposes.

Feature 4: Reminder Notifications

Sends automated alerts reminding users of upcoming trips. Allows customization of notification times and frequencies. Ensures timely delivery even during peak traffic periods. Feature 5: Profile Management Tool

Stores personal information securely for streamlined rebooking processes. Updates existing entries automatically based on recent activity. Protects sensitive data against unauthorized access through encryption techniques. End of Document.

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